

Math 210B: Homework 2

Lance Remigio

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Problem 1 (Spring 2022 Ring Problem 1). Let R be a commutative ring with identity 1. An element e of R is called **idempotent** if $e^2 = e$. Let e and f be nonzero idempotents of R with $e + f = 1$ and $ef = 0$.

(a) Show that the ideals (e) and (f) of R are in fact rings with identity different than 1.

Proof. We show that (e) does indeed define a ring. To do this, we want to show that (e) is a subring of R and so the ring structure of (e) will follow. First, we show that (e) is closed under addition. Let $x, y \in (e)$. Then $x = r_1e$ and $y = r_2e$ for some r_1, r_2 in R . Then

$$\begin{aligned}x + (-y) &= r_1e + (-r_2e) \\&= r_1e - r_2e \\&= (r_1 - r_2)e.\end{aligned}$$

Since R is a ring $r_1 - r_2 \in R$. Hence, $x + (-y) \in (e)$. Now, we want to show that (e) is closed under multiplication. Let $x, y \in (e)$. Then for some r_1 and r_2 in R , it follows that $x = r_1e$ and $y = r_2e$. Since R is commutative and $e^2 = e$, it follows that

$$\begin{aligned}xy &= (r_1e)(r_2e) = (r_1r_2)e^2 \\&= (r_1r_2)e.\end{aligned}$$

Since R is closed under multiplication, it follows that $r_1r_2 \in R$ and so $xy \in (e)$. Thus, we have that (e) is a ring. Showing that (f) is a ring follows similarly with f in place of e . We claim that the identity for (e) and (f) will be $1 + f$ and $1 + e$, respectively. Indeed, if $x \in (e)$ is arbitrary, then $x = re$ for some $r \in R$. Then

$$(1 + f)x = (1 + f)(re) = re + f(re) = re + r(ef) = re + 0 = re.$$

Since R is a commutative ring, it follows that $(1 + f)x = x(1 + f)$. Thus, we get that

$$(1 + e)x = x(1 + e) = x.$$

Similarly, we see that if $x \in (f)$ is arbitrary, then $x = rf$ for some $r \in R$. Then

$$(1 + e)x = (1 + e)(rf) = rf + e(rf) = re + r(ef) = rf + 0 = rf.$$

Since R is a commutative ring, it follows that $(1 + e)x = x(1 + e)$. ■

(b) Prove that $(e) \times (f)$ and R are isomorphic.

Proof. Define $\varphi : R \rightarrow (e) \times (f)$ by $\varphi(r) = (re, rf)$. First, we show that φ is a ring homomorphism. Let $r_1, r_2 \in R$. Then

$$\begin{aligned}\varphi(r_1 + r_2) &= ((r_1 + r_2)e, (r_1 + r_2)f) \\&= (r_1e + r_2e, r_1f + r_2f) \\&= (r_1e, r_1f) + (r_2e, r_2f) \\&= \varphi(r_1e, r_1f) + \varphi(r_2e, r_2f).\end{aligned}$$

Hence, φ is additive. Now, we show that φ is multiplicative. Then

$$\begin{aligned}\varphi(r_1 r_2) &= ((r_1 r_2)e, (r_1 r_2)f) \\ &= ((r_1 r_2)e^2, (r_1 r_2)f^2) \\ &= ((r_1 e)(r_2 e)(r_1 f)(r_2 f)) \\ &= (r_1 e, r_1 f) \cdot (r_2 e, r_2 f) \\ &= \varphi(r_1) \cdot \varphi(r_2).\end{aligned}$$

Note that since f and e are nonzero idempotents, it follows that

$$\begin{aligned}\ker(\varphi) &= \{r \in R : \varphi(r) = 0_{(e) \times (f)}\} \\ &= \{r \in R : (re, rf) = (0_R, 0_R)\} \\ &= \{r \in R : re = 0_R \text{ and } rf = 0_R\} \\ &= \{r \in R : r = 0_R\} \\ &= \{0_R\}.\end{aligned}$$

Hence, it follows that φ is injective. Now, we show that φ is surjective. Let $(r_1 e, r_2 f) \in (e) \times (f)$. Since $ef = 0_R$ and $e + f = 1_R$, define $r = r_1 f + r_2 f$. Then we have

$$\begin{aligned}\varphi(r) &= \varphi(r_1 f + r_2 f) \\ &= ((r_1 e + r_2 f)e, (r_1 e + r_2 f)f) \\ &= (r_1 e^2 + r_2(fe), r_1(e f) + r_2 f^2) \\ &= (r_1 e + 0, 0 + r_2 f) \\ &= (r_1 e, r_2 f).\end{aligned}$$

Hence, it follows that φ is surjective. Now, we conclude that φ is bijective homomorphism and so $(e) \times (f) \cong R$. ■

Problem 2 (Spring 2020 Rings Problem 1). Let R be a commutative ring with unity, and let P be a prime ideal in R .

- (a) If I and J are ideals in R with $I \cap J \subseteq P$, show that one of I or J is a subset of P .

Proof. We want to show that either $I \subseteq P$ or $J \subseteq P$. Let $x \in I$ and $y \in J$ be arbitrary. Now, since I and J are ideals (and the fact that R is commutative), it follows that both $xy \in I$ and $yx \in J$ where $xy = yx$. Hence, $xy = yx \in I \cap J$. Since $I \cap J \subseteq P$, it follows that $xy \in P$. But this tells us that either $x \in P$ or $y \in P$ since P is a prime ideal. This tells us that either $I \subseteq P$ or $J \subseteq P$. ■

- (b) If R is finite, then explain why R/P .

Solution. Since R is finite and P is a prime ideal, it follows that R/P is a finite integral domain. Thus, we see that R/P must be a field by corollary to proposition 2. ■

Problem 3 (Problem 5 Section 7.4). Prove that if M is an ideal such that R/M is a field then M is maximal ideal.

Proof. Our goal is to show that M is a maximal ideal; that is, we want to show that for all ideals $K \leq R$, if $M \leq K \leq R$, then either $K = M$ or $K = R$. Let $K \leq R$ be an arbitrary ideal. Since M is an ideal and R/M is a field, it follows that R/M is a commutative division ring by definition. Thus, the only ideals of R/M are $\bar{0}$ and R/M itself. Hence, $K/M = \bar{0}$ or $K/M = R/M$. Thus, we have $K = M$ or $K = R$. Thus, M is maximal in R . ■

Problem 4 (Problem 7 Section 7.4). Let R be a commutative ring with 1. Prove that the principal ideal generated by x in the polynomial ring $R[x]$ is a prime ideal if and only if R is an integral domain. Prove that (x) is a maximal ideal if and only if R is a field.

Proof. Since (x) is a prime ideal and $(x) \subseteq R[x]$, it follows that $R[x]/(x)$ is an integral domain. To show that R is an integral domain, it suffices to show that

$$R[x]/(x) \cong R$$

via the first isomorphism theorem for rings. Define $\varphi : R[x] \rightarrow R$ by $\varphi(f(x)) = a_0$ where

$$f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n.$$

We will show that φ is a ring homomorphism, $(x) = \ker(\varphi)$, and $\varphi(R[x]) = R$. First, we show that φ is additive and multiplicative. Indeed, for any $f(x), g(x) \in R[x]$, we have that

$$\varphi(f(x) + g(x)) = a_0 + b_0 = \varphi(f(x)) + \varphi(g(x)).$$

where $f(x)$ is defined as before and

$$g(x) = b_0 + b_1x + b_2x^2 + \cdots + b_nx^n.$$

Since multiplication is defined in the following way for polynomials

$$\sum_{i=0}^k a_i b_{k-i},$$

it follows that

$$\varphi(f(x)g(x)) = a_0b_0 = \varphi(f(x))\varphi(g(x)).$$

Thus, we conclude that φ is a ring homomorphism. Now, we show that $\ker(\varphi) = (x)$. Indeed, we see that

$$\begin{aligned} \ker(\varphi) &= \{f(x) \in R[x] : \varphi(f(x)) = 0_R\} \\ &= \{f(x) \in R[x] : a_0 = 0_R\} \\ &= \{f(x) \in R[x] : a_1x + a_2x^2 + \cdots + a_nx^n\} \\ &= \{f(x) \in R[x] : x(a_1 + a_2x + \cdots + a_nx^{n-1})\} \\ &= \{f(x) \in R[x] : xg(x) \text{ for some } g(x) \in R[x]\} \\ &= (x). \end{aligned}$$

It is relatively straightforward to see that $\varphi(R[x]) = R$ because for any $r \in R$, we can define a polynomial $f(x) = r$ to be the constant polynomial and so $\varphi(f(x)) = r$. Thus, the first isomorphism theorem tells us that

$$R[x]/\ker(\varphi) \cong \varphi(R[x]) \implies R[x]/(x) \cong R.$$

Now, since $R[x]/(x)$ is an integral domain, it also follows that R is an integral domain. That is, (x) is a prime ideal if and only if $R[x]/(x)$ is a field if and only if R is an integral domain. ■

Problem 5 (Problem 37). A commutative ring R is called a **local ring** if it has a unique maximal ideal. Prove that if R is a local ring with maximal ideal M , then every element of $R - M$ is a unit. Prove conversely that if R is a commutative ring with 1 in which the set of nonunits forms an ideal M , then R is a local ring with unique maximal ideal M .

Proof. (My attempt) Let $r \in R - M$. Our goal is to show that r is a unit; that is, there exists an $s \in R$ such that $rs = sr = 1_R$. Since M is maximal ideal and R is a commutative ring, $M \subseteq R$, we have that R/M is a field, and so R/M is a commutative division ring. That is, there exists

$s + M \in R/M$ such that

$$(r + M)(s + M) = 1_R + M.$$

Note that since $r \in R - M$, we have that $r \notin M$ and so we have that $r + M \neq 0_R + M$. Now, we get that

$$\begin{aligned}(r + M)(s + M) &= 1_R + M \\ (rs) + M &= 1_R + M.\end{aligned}$$

Thus, we get that $rs - 1_R \in M$. **Now, I wanted to show that $rs - 1_R = 0_R$, but I wasn't sure how to go about doing that.**

($\Leftarrow$) Suppose that M is the ideal containing all of the non-units of R . Our goal is to show that R is a local ring; that is, R contains a unique maximal ideal. We claim that M is the unique maximal ideal we are looking for. First, we show that M is maximal in R . To do this, we need to show that for all $K \leq R$, if $M \leq K \leq R$, then either $K = M$ or $M = R$. To this end, let $K \leq R$ be any ideal in R . Let $x \in K$. Then either x is a unit or x is a non-unit of R . If x is a unit, then $x \in R$ since $K \leq R$. Since R is a commutative ring, it follows that $K = R$. If x is a non-unit, then $x \in M$ and so $K \leq M$. But we already assumed that $M \leq K$ and so we conclude that $M = K$. Thus, we find that M is indeed a maximal ideal of R .

To show that M is the unique ideal, it suffices to consider another maximal ideal M' that contains the non-units of R . Since M' is maximal, we have (by definition) $M \leq M' \leq R$, then either $M = M'$ or $M' = R$. If $M = M'$, then M is unique. If $M' = R$, then $M \leq M'$, but since M is maximal it follows that $M = M'$, implying that M is unique. Hence, in either case, we find that M is the unique maximal ideal of R . ■

Proof. (A proof of the forwards direction from the internet) Let $r \in R - M$ be arbitrary. Suppose for contradiction that r is NOT a unit. Then $(r) \neq R$ and so let M' be a maximal ideal containing (r) via Zorn's Lemma. Since R only maximal ideal, $M = M'$. But then this would imply that $r \in M$ and so a contradiction. Thus, r must be a unit. ■